

Code-verified novelty of RadCluster

Revised consistency analysis against RadCluster.2.1 — September 14, 2026

1 Production bias and simultaneous loop–cavity evolution

The supplied analysis is correct that the cascade source creates a strong partition bias. For $x = i, v$,

$$G_1^x = \eta \dot{d}(1 - f_{\text{cl}}^x), \quad G_n^x = \eta \dot{d} A_x n^{-s_x}, \quad \sum_n n G_n^x = \eta \dot{d}.$$

Thus equal total SIA and vacancy inventories are created, but assigned to different mobility classes. For the fission spectrum:

Measure	SIA	vacancy	ratio
clustered atom fraction	0.580	0.150	3.87 (SIA/V)
cluster-number fraction	0.1086	0.0485	2.24 (SIA/V)
monomer atom fraction	0.420	0.850	2.02 (V/SIA)
mean produced cluster size	5.34	3.09	—

The factor of two is therefore appropriate for produced cluster number or inverse monomer fractions; the clustered atom partition differs by about 3.9.

Cascade-born SIA clusters seed loops, while the larger vacancy-monomer source feeds cavities. RadCluster evolves both ladders with SIA–SIA and V–V coalescence, residual-size reactions

$$I_n + V_m \rightarrow I_{n-m}, \quad V_{m-n}, \quad \text{or } 0,$$

cavity shrinkage, emission, helium processes and external sinks. This size-resolved coupled evolution is the main advance beyond a two-monomer swelling model.

The original transport interpretation is too categorical. The implemented mixed-dimensional cavity kernel contains

$$\mathcal{K}_{n,m}^{\text{cav}} \propto \frac{m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \widehat{L}^2 m^{-1/3}}.$$

Suppression values quoted without m apply only to the $m = 1$ reference; larger cavities are less suppressed. One should say that dimensionality changes the relative rates of cavity interception, coalescence and sink capture, not that 1D clusters cannot find cavities. Coalescence reduces mobile cluster number and may transfer SIA inventory into large or sessile loops, but it conserves SIA mass.

For balanced production and initial inventories, the correct signed balance is

$$S_V(t) - S_I(t) = J_I^{\text{sink}}(t) - J_V^{\text{sink}}(t), \tag{1}$$

up to numerical residuals and boundary losses. Internal coalescence and mutual annihilation cancel from this balance.

Equal capture efficiencies do not make the right side zero. Setting $Z_i = Z_v$ removes preferential elastic capture, but the sink fluxes can remain unequal because mobile concentrations and diffusivities differ precisely because of production bias. Hence equal Z does not force $S_V = S_I$.

The decisive numerical design is

	$f_{\text{cl}}^i = f_{\text{cl}}^v$	$f_{\text{cl}}^i > f_{\text{cl}}^v$
$Z_i = Z_v$	reference	production bias only
$Z_i \neq Z_v$	sink bias only	combined.

Equalize dislocation, grain-boundary and precipitate capture-efficiency factors while fixing sink densities, transport, binding, helium conditions and dose rate. The production-only case supports the claim if both loop and cavity inventories and mean sizes grow relative to the unbiased-source reference. Report source, coalescence, annihilation, cavity-capture and external-sink fluxes with Eq. (1).

Golubov–Singh–Trinkaas rate theory already permits production-biased swelling. RadCluster’s stronger novelty is simultaneous size-resolved CD evolution of both cluster families with mixed-dimensional transport and nonlocal reactions.

2 Bin moments versus Kiritani and Golubov

RadCluster stores $\mu_k^{(p)} = \sum_{n \in \mathcal{B}_k} n^p c_n$ and implements

$$\{\mu_k^{(p)}\} \xrightarrow{R} c_n^{\text{rec}} \xrightarrow{E} c_n^{\text{full}} \xrightarrow{P} \dot{\mu}_k^{(p)} = \sum_{n \in \mathcal{B}_k} n^p \dot{c}_n^{\text{full}}.$$

This reconstruct–evaluate–project method avoids deriving a separate grouped operator for every reaction family. Intra-bin flux is not absent: per-size transfers are evaluated and cancel from a projected moment only if their net moment contribution is zero.

	Kiritani-type	Golubov improved	RadCluster
State	uniform group density	two variables; linear profile	constant, linear or log-normal ($P = 1, 2, 3$)
Operator	grouped equation	derived group equations and mean flux	reconstruct grid, evaluate per-size RHS, project
Coefficients	group approximations	grouped coefficients	original size-dependent coefficients on reconstruction
Nonlocal reactions	special treatment	grouped extensions needed	same machinery handles coalescence and residual reactions
Conservation	may lose consistency	can conserve selected moments	inherits projected stoichiometry with qualifications
Work	group-level	group-level	reduced ODE state but full-grid sweeps

Golubov’s method is not intrinsically nonconservative; its two-variable scheme improves number/content balances over uniform grouping. RadCluster’s $P = 2$ closure is analogous to a linear two-moment representation, but its dynamical operator is assembled differently.

Exact projection preserves stoichiometric first moments, but conservation is not unconditional: linear reconstruction clips negative values; the discrete log-normal closure exactly normalizes $\mu^{(0)}$ but only approximately recovers higher moments; the upper boundary can lose products; and floors and solver tolerances enter the residual. The unverified $O((r-1)^2)$, $O((r-1)^3)$ and $O((r-1)^4)$ orders should be removed unless derived for the implemented discrete closures. The proper claim is *stoichiometrically conservative at the projected-reaction level and continuously audited*. R–E–P reduces solver-state dimension but still sweeps $I + V$; its advantage is reaction fidelity, not the full work reduction of a strict grouped operator.

Recommended novelty statement

RadCluster is a production-spectrum-resolved CD framework coupling asymmetric cascade partitioning, mixed 1D/3D SIA transport, SIA–SIA and vacancy–vacancy coalescence, residual-size SIA–vacancy reactions, cavity capture/emission, helium coupling, and sink accounting. Controlled calculations with equal capture efficiencies isolate simultaneous loop and cavity evolution caused by production bias rather than preferential sink capture. Its reconstruct–evaluate–project bin-moment reduction retains the original size-dependent kernels and projected stoichiometric balances without reaction-specific inter-group operators, while auditing reconstruction, boundary and solver errors.

Code anchors

[Production](#); [bin moments](#); [kernels](#); [formulation](#). Golubov, Singh and Trinkaus, *J. Nucl. Mater.* (2000), doi:[10.1016/S0022-3115\(99\)00171-3](#).